

Rahul Khanchandani

Senior Software Engineer — Full Stack

React / Next.js / Node.js / TypeScript

Ahmedabad, India | +91 7984 137059 | rahul.19797@gmail.com

LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Senior Full Stack Engineer with 6+ years of experience building scalable, high-performance web applications across frontend and backend technologies. Expertise in React, Next.js, TypeScript, Node.js, and modern web architectures, with hands-on experience designing REST APIs, real-time WebSocket systems, DynamoDB data models, Redis-based authentication, and asynchronous processing using AWS SQS.

Strong background in frontend performance optimization, achieving Lighthouse scores of 85+ and improving Core Web Vitals through SSR, SSG, ISR, code splitting, virtualization, lazy loading, and caching strategies. Experienced in building analytics dashboards, reporting systems, RBAC modules, real-time notifications, and automated workflows. Skilled in system design, structured logging, CI/CD pipelines, testing (Jest, React Testing Library, TDD), accessibility (WCAG 2.1 AA), and mentoring engineers. Comfortable owning end-to-end product development, making architectural decisions, and leveraging AI-assisted development tools to accelerate delivery and improve engineering productivity.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), HTML5, CSS3, SCSS, SQL

Frontend: React.js, Next.js, React Hooks, Redux Toolkit, React Query, React-Flow, react-window, Material-UI, Tailwind CSS, Nivo Charts, Storybook

Backend: Node.js, Express.js, REST APIs, GraphQL, WebSocket, JWT Authentication, Redis Session Management, MySQL (Joins, Aggregations), DynamoDB (GSI, Key Conditions, Filter Expressions), AWS SQS, Puppeteer, Cron Jobs, Pino Logging

Architecture & System Design: SSR, SSG, ISR, Micro-Frontend (Module Federation), Monorepo, Frontend & Backend System Design, Real-Time Systems, Queue-Based Processing, Caching Strategies

Performance: Core Web Vitals (LCP, CLS, INP), Lighthouse Optimization, Code Splitting, Lazy Loading, Virtualization, Memoization, Intersection Observer, Image Optimization

Testing: Jest, React Testing Library, TDD, Unit Testing, Integration Testing, 90%+ Test Coverage

DevOps & Tooling: Git, GitHub Actions, CI/CD, Webpack, Vite, Sentry, Chrome Extension API, Figma

Cloud & Infrastructure: AWS SQS, EC2, CloudWatch

Security & Accessibility: RBAC, JWT, Redis Authentication, WCAG 2.1 AA

AI Tools: Claude AI, OpenAI Codex, AI-Assisted Development

Methodologies: Agile, Scrum, Code Reviews, Technical Mentorship

PROFESSIONAL EXPERIENCE

Senior Software Engineer · AppVentory

Sep 2023 – Present

Ahmedabad, India

- Architected and owned the end-to-end frontend of an affiliate link tracking platform, building a real-time analytics dashboard and fraud detection UI processing high-volume daily events with <200ms latency and directly supporting a new revenue stream.
- Built Node.js backend APIs to aggregate and serve real-time analytics data for leads and deal conversion dashboards; applied parallel data processing and database indexing to reduce server load and improve query response times.
- Modeled and queried DynamoDB tables for analytics data using partition-key queries, key condition and filter expressions, and Global Secondary Indexes (GSIs) for efficient lookups beyond the primary access pattern.
- Implemented centralized structured JSON logging using Pino across backend services, capturing request-level audit logs and module-level event/error logging to improve traceability and debugging efficiency.
- Automated PDF report generation using Puppeteer to render and export data directly from live frontend dashboard pages, with reports delivered automatically via email.
- Identified synchronous PDF generation as a CPU-heavy operation blocking the API response cycle and introduced an asynchronous AWS SQS architecture. CloudWatch metrics showed CPU utilization dropping from approximately 90% to 30% during peak report generation, improving resilience and scalability.
- Built and optimized Next.js applications using SSR, SSG, and ISR rendering strategies, achieving Lighthouse scores of 85+ and strong Core Web Vitals through code splitting, lazy loading, and image optimization.
- Built server-side WebSocket services and integrated GraphQL subscriptions to power real-time notifications and live dashboard updates, reducing over-fetching by 30%.

- Achieved 90%+ unit and integration test coverage on critical business logic using Jest and React Testing Library with TDD, contributing to a 40% reduction in production bugs.
- Set up and maintained GitHub Actions CI/CD pipelines for automated testing, linting, and deployment across multiple product verticals.
- Mentored 2 junior engineers through structured code reviews, pair programming, and architectural guidance.

Stack: TypeScript, Next.js, Node.js, Express.js, React-Flow, React Query, WebSocket, GraphQL, DynamoDB, AWS SQS, Puppeteer, Cron Scheduling, Pino, Database Indexing, Storybook, Jest, GitHub Actions, Module Federation, Claude AI, OpenAI Codex, Chrome Extension API, Figma

Software Engineer · Magenta Connect Pvt Ltd

Jul 2021 – Sep 2023

Ahmedabad, India

- Modernized a legacy jQuery/JavaScript codebase to React 16+ with Hooks, Redux Toolkit, and TypeScript, improving overall frontend performance by 40% and significantly reducing bundle size.
- Built a Kanban-based ticketing system, Role-Based Access Control (RBAC) module, and User Session Management supporting 500+ concurrent users without degradation.
- Implemented Redis-backed session authentication by storing and validating JWT tokens on every request, enabling fast and reliable session management at scale.
- Diagnosed a production performance issue where a data-heavy screen became unresponsive for users with large record sets. Traced the root cause to thousands of DOM elements being rendered simultaneously and implemented react-window virtualization, reducing DOM load and improving scroll and initial rendering performance.
- Developed interactive KPI dashboards using Nivo Charts and optimized API calls using Intersection Observer-based lazy fetching, eliminating unnecessary network requests during page load.
- Wrote MySQL JOIN and aggregation queries to compute sales statistics and enrich sales records with product information, powering dashboard APIs.
- Integrated Sentry error monitoring with custom alerting workflows, reducing Mean Time to Detection (MTTD) for critical frontend bugs by 50%.
- Championed Test-Driven Development using Jest, reducing regression issues by 35%.
- Served as a core code reviewer and frontend subject matter expert, improving code quality and architectural consistency.

Stack: TypeScript, React, Node.js, Redux Toolkit, MySQL, Redis, JWT, Nivo Charts, Sentry, Jest, react-window, react-intersection-observer, Figma

Frontend Engineer · Lucent Innovation

Aug 2020 – Jun 2021

Ahmedabad, India (Remote)

- Optimized e-commerce frontend performance, improving Lighthouse scores from 55 to 80+ and improving Core Web Vitals through code splitting, lazy loading, and asset optimization.
- Implemented advanced UX patterns including debounced search, infinite scroll with throttling, and skeleton loading states to improve perceived performance and user experience.
- Developed fully responsive, mobile-first websites using CSS Media Queries, ensuring consistent UI adaptation across mobile, tablet, and desktop breakpoints.
- Reduced redundant network calls by introducing memoization and client-side caching strategies for API responses.
- Improved cross-browser compatibility across Chrome, Firefox, Safari, and Edge, ensuring consistent UI rendering and performance.

Stack: JavaScript, React, jQuery, SCSS, Material-UI, CSS Media Queries

EDUCATION

MSc (IT) · DAIICT

Jul 2018 – May 2020

Gandhinagar, India

BCA · Gujarat University

Jul 2015 – May 2018

Ahmedabad, India